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Lecture due Jan 24, 2024 20:30 IST

Formula for the dimension of the nullspace. Suppose that the result of putting a matrix **A** in row echelon form is **B**. Then **NS (A) = NS (B)** (since row reductions do not change the solutions), and

$$\dim \text{NS}(\mathbf{A}) = \# \text{ non-pivot columns of } \mathbf{B}.$$

(The boxed formula holds since it is the same as $\dim \text{NS}(\mathbf{B}) = \# \text{ free variables}$.)

Steps to find a basis and the dimension of the space of solutions to a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$:

1. Perform Gaussian (Gauss–Jordan) elimination on **A** to convert it to a matrix **B** in (reduced) row echelon form.
2. Identify the pivots of **B**.
3. Count the number of **non-pivot** columns of **B**; that number is $\dim \text{NS}(\mathbf{A})$.
4. Use back-substitution to find the general solution to $\mathbf{Bx} = \mathbf{0}$.
5. The general solution will be expressed as the general linear combination of a list of vectors; that list is a basis for **NS (A)**.

Warning: You must put the matrix in row echelon form before counting non-pivot columns!

Nullspace concept check I

1/1 point (graded)
Which of the following is a basis for the nullspace of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}?$$

- ☐ $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$
- ☐ $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$
- ☒ $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$
- ☐ $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$



Solution:

The matrix is already in row echelon form. It has 2 pivots (in orange), therefore 2 pivot variables, and one free variable corresponding to the non-pivot column in blue:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use back substitution to get the general solution $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ for the nullspace. The matrix has one non-pivot column, so we expect a 1-dimensional nullspace. We start the back substitution with

$x_3 = 0.$

The variable x_2 is a free variable, so we set

$x_2 = c$ for a parameter c

Then we have from the first row

$x_1 = 0.$

The general solution is

$$\mathbf{x} = \begin{pmatrix} 0 \\ c \\ 0 \end{pmatrix} = c \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

so a basis for the nullspace of $\mathbf{A}$ is $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$

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You have used 1 of 3 attempts

i Answers are displayed within the problem

Nullspace concept check II

1/1 point (graded)

What is the nullspace of the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$?

- ☐ The x -axis in $\mathbb{R}^3$.
- ☐ The xy -plane in $\mathbb{R}^3$.
- ☐ The xz -plane in $\mathbb{R}^3$.
- ☒ The yz -plane in $\mathbb{R}^3$.
- ☐ All of $\mathbb{R}^3$.



Solution:

The nullspace of $\mathbf{A}$ is the yz -plane in $\mathbb{R}^3$.

The matrix is already in row echelon form. It has 1 pivot (in orange), therefore one pivot variable x , and two free variables y and z corresponding to the non-pivot columns in blue.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

To find the nullspace, we set the free variables equal to parameters:

$y = c_1$

$z = c_2$

Then the equation $\mathbf{Ax} = \mathbf{0}$ forces $x = 0$. Thus the nullspace is all vectors of the form $\begin{pmatrix} 0 \\ c_1 \\ c_2 \end{pmatrix}$ for c_1 and c_2 any real numbers. This is exactly the yz -plane.

Alternative solution: Notice that in this case, we can see that system $\mathbf{Ax} = \mathbf{0}$ has one equation, $x = 0$. This forces $x = 0$ and y and z can be anything. This is exactly a description of the yz -plane.

Calculator

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To see this we notice that

$$A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

as well as that

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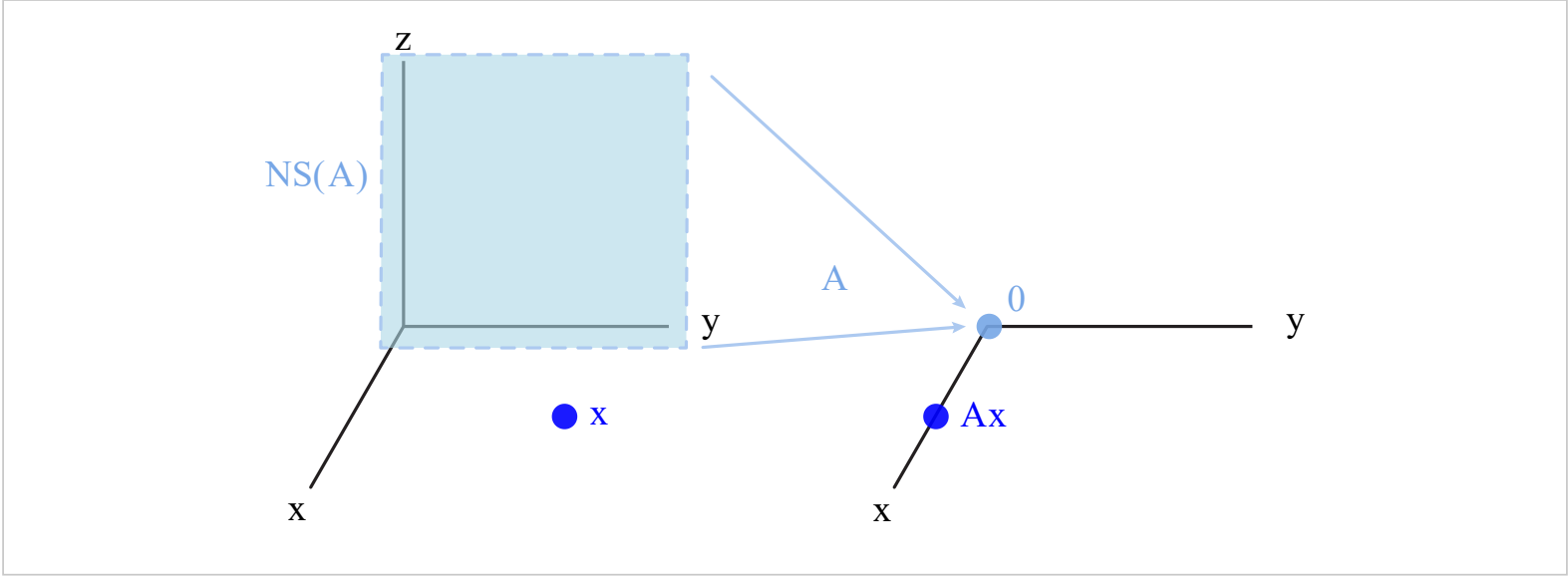
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with the x component equal to zero: this is the yz -plane.



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